

How Stable Are Regression Conclusions Across Subpopulations? A Finite-Sample Empirical Study

Draft planning document

July 14, 2026

1 Introduction

Applied researchers routinely fit a single linear regression model to data pooled across naturally occurring subgroups – age cohorts, regions, treatment arms, demographic categories – and interpret the resulting coefficients as general descriptions of the relationship between predictors and response. This practice is common because pooled models use all available data, but it rests on the implicit assumption that the regression relationships are sufficiently similar across groups. When that assumption fails, the pooled coefficients may be poor summaries for any subgroup individually, and conclusions about sign, magnitude, and predictive accuracy may not hold outside the sample used to fit the model.

The theoretical basis for this concern is well established. Berk et al. [?] show that OLS coefficients, under misspecification, are best linear approximations to the true conditional mean function, and that these approximations depend on the marginal distribution of the regressors – not only on the response surface. When subgroups differ in their predictor distributions, the pooled estimand itself changes, even if the true data-generating process is the same within each subgroup. Coefficient heterogeneity across groups can therefore arise from distributional differences alone, independently of any difference in underlying relationships. A well-known extreme case is Simpson’s paradox, in which a pooled association reverses direction within every subgroup because the pooled estimate conflates a within-group slope with a between-group compositional effect.

A substantial literature has developed tools for detecting coefficient instability, but most of it is focused on time-ordered data. The structural break tests of Chow (1960) and the CUSUM-based procedures reviewed by Garbade [?] are designed to detect discrete or gradual changes as observations accumulate over time. Watson and Engle [?] extend this to a stationary AR(1) alternative for time-varying coefficients. These tests are powerful within their domain but assume a specific temporal structure – a natural ordering of observations and a structured class of alternatives – that does not apply to cross-sectional subgroup comparisons where groups are defined by observed characteristics.

A separate literature on heterogeneous treatment effects asks whether a treatment impact differs across subgroups, using methods such as interaction regression, propensity score stratification, and causal forests [?]. Tsoi and Sun [?] conduct a recent Monte Carlo comparison of interaction and stratified regression for detecting subgroup differences in treatment effects, finding that the two approaches can yield conflicting conclusions depending on sample size, subgroup balance, and covariate correlation. Their results highlight the methodological risk of mixing the two approaches in reporting. That literature, however, focuses on a single binary treatment variable and frames stability as a hypothesis-testing question – whether a subgroup difference is statistically detectable

– rather than asking how large the differences are, whether effect directions change, or when a model trained on one subgroup predicts poorly for another.

Two other lines of work approach subgroup stability from different angles. Bertsimas and Paskov [?] propose enforcing stability by optimally selecting training and validation sets to minimize worst-case prediction error across subpopulations. Their robust optimization framework builds stability in by design and produces coefficients with lower variance and better support recovery than random splits. In panel econometrics, Liu et al. [?] develop an M-estimation approach that identifies and estimates latent group structure when subgroup membership is unknown and coefficients are heterogeneous across groups. Both papers offer solutions for settings where stability must be constructed or group membership must be recovered. Neither addresses the common diagnostic case in which groups are observed and the analyst wants to assess how stable the pooled regression conclusions already are.

This paper addresses that gap. We treat group membership as observed and ask, using a small set of interpretable metrics, how stable the conclusions of a pooled linear regression are across subpopulations in finite samples. The metrics capture four dimensions of stability: whether coefficient signs agree across groups, how much coefficient magnitudes drift relative to the pooled estimate, whether that drift is explained by differences in covariate distributions, and how much predictive accuracy degrades when a model trained on one subgroup is applied to another. We study these metrics in a controlled simulation that varies sample size, subgroup balance, predictor correlation, and the type of heterogeneity present, and we illustrate with a real-data case study using NHANES data. The goal is not to propose a new estimator but to provide practical diagnostic guidance: when should a pooled regression conclusion be treated with caution, and what does that caution look like in finite samples?

The remainder of the paper is organized as follows. Section 2 defines the stability metrics and regression models compared. Section 3 presents the simulation study. Section 4 presents the NHANES case study. Section 5 discusses implications and limitations.

2 Methods

We consider a response Y , predictors $X \in \mathbb{R}^p$, and subgroups indexed by $g \in \{1, \dots, G\}$. Let n_g denote the sample size in subgroup g , and let $\hat{\beta}_{j,g}$ denote the estimated coefficient for predictor j in subgroup g . The pooled estimate is denoted by $\hat{\beta}_{j,\text{pooled}}$.

The goal of the Methods section is not to introduce a new estimator. Instead, it sets up a simple comparison framework for asking when a pooled linear regression remains stable across subpopulations, and when it breaks down. We treat the pooled model as the baseline and then examine subgroup-specific departures using five quantities: sign stability, standardized coefficient drift, context-normalized drift, transported prediction gap, and a summary of covariate shift.

2.1 Core Stability Metrics

Table 1 summarizes the five primary metrics used in the paper.

In SSR_j , each pair (a_m, b_m) is a selected subgroup or pooled-vs-subgroup comparison.

Higher values of D_j^* , $\tilde{D}_j$, $\text{TPG}_{a \rightarrow b}$, and $\text{CSS}_{a,b}$ indicate greater instability or mismatch, while higher SSR_j indicates greater directional stability.

Table 1: Core stability metrics used in the study

Metric	Formula	Interpretation
Sign stability rate (predictor j)	$I_{j,m} = \mathbf{1}(\text{sgn } \hat{\beta}_{j,a_m} = \text{sgn } \hat{\beta}_{j,b_m}),$ $\text{SSR}_j = \frac{1}{M} \sum_{m=1}^M I_{j,m}.$	Proportion of comparisons with the same effect direction.
Standardized coefficient drift (predictor j)	$D_j = \max_g \hat{\beta}_{j,g} - \hat{\beta}_{j,\text{pooled}} ,$ $D_j^* = D_j / \widehat{\text{SE}}(\hat{\beta}_{j,\text{pooled}}).$	Largest subgroup departure from pooled effect, scaled for cross-predictor comparability.
Context-normalized drift (predictor j)	$M_j = \frac{1}{\binom{G}{2}} \sum_{a < b} \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}},$ $\tilde{D}_j = \frac{D_j^*}{1 + M_j}.$	Coefficient drift scaled by subgroup mean separation, so large drift is interpreted relative to observed covariate spread.
Transported prediction gap ($a \rightarrow b$)	$\text{TPG}_{a \rightarrow b} = \text{RMSE}_{a \rightarrow b} - \text{RMSE}_{a \rightarrow a}$ (analogously for MAE).	Out-of-group performance loss relative to in-group performance.
Covariate shift summary between groups a and b	$\text{CSS}_{a,b} = \frac{1}{p} \sum_{j=1}^p \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}}.$	Average standardized difference in predictor distributions across groups.

2.2 Regression Models Compared

For each dataset or simulated replicate, we compare four linear regression specifications. The first is the pooled model,

$$Y = \beta_0 + X^\top \beta + \varepsilon,$$

which imposes a single coefficient vector across all subgroups. The second adds a group indicator,

$$Y = \beta_0 + X^\top \beta + \gamma G + \varepsilon,$$

which allows subgroup-specific intercept shifts but keeps slopes common. The third adds interactions,

$$Y = \beta_0 + X^\top \beta + \gamma G + X^\top \delta G + \varepsilon,$$

which allows the slope coefficients to vary by subgroup. The fourth fits separate subgroup-specific OLS models, so that each group has its own intercept and slope vector.

This comparison is intentionally simple. It lets us isolate whether instability arises from the pooled structure itself, from slope heterogeneity, or from changes in the predictor distribution across groups.

2.3 Simulation Design

The simulation study begins from a pooled baseline and then introduces subgroup structure as a stress test. We vary four main ingredients: subgroup balance, sample size, predictor correlation, and the type of heterogeneity present. For subgroup balance, we consider both roughly equal group sizes and imbalanced settings where one group is much smaller than the others. For sample size, we use small and moderate samples to see how much apparent instability is driven by finite-sample noise. For predictor correlation, we vary the degree of collinearity so that coefficient estimates are not artificially easy to separate.

The baseline scenario is generated from a single global linear model, so all subgroups follow the same regression rule. The remaining scenarios add subgroup structure by allowing predictor

distributions, coefficients, or both to vary by group. This lets us compare a setting where the pooled model should work well against settings where the pooled model is only an approximation.

We study four scenarios. In the no-heterogeneity scenario, all groups share the same regression coefficients, predictor distribution, and noise level. In the covariate-shift scenario, the coefficients stay fixed but the predictor distribution changes across groups. In the coefficient-heterogeneity scenario, the coefficients differ across groups while the predictor distribution is held fixed. In the combined scenario, both the coefficients and the predictor distribution vary across groups.

For each replicate, we generate the data, fit all four models, and record coefficient estimates, standard errors, signs, and prediction errors. Prediction is evaluated both within-group and across groups. This is important because the paper is not only asking whether coefficients change, but also whether a pooled model that appears acceptable overall continues to predict well in each subgroup.

The simulation outputs are summarized at the replicate level and then aggregated across replicates to compute the stability metrics above. This makes it possible to compare how often sign reversals occur, how far subgroup coefficients drift from the pooled estimate, and how much predictive error increases under transport.

2.4 Real-Data Workflow

The real-data analysis follows the same logic as the simulations. For each dataset, we identify a continuous response, a clear subgroup variable, and a moderate set of predictors. After basic cleaning and preprocessing, we first fit the pooled regression as the baseline. We then fit the same four regression specifications used in the simulation study and compare pooled, interaction-based, and subgroup-specific estimates, along with in-group and out-of-group prediction performance.

The purpose of the real-data section is not to claim universal generalization. Rather, it is to show how the stability metrics behave in applied settings where subgroup structure is present but not controlled by the analyst. The real-data examples should therefore be interpreted as illustrations of the same mechanisms studied in simulation: coefficient instability, covariate shift, and transport failure.

2.5 How the Metrics Are Used

The five metrics play different roles in the analysis. Sign stability is the most direct measure of whether substantive conclusions reverse across subgroups. Standardized coefficient drift measures how far subgroup estimates move away from the pooled estimate on a comparable scale. Context-normalized drift scales that movement by observed subgroup covariate separation. The transported prediction gap measures whether predictive performance deteriorates when a pooled model or subgroup model is moved from one group to another. Finally, the covariate shift summary provides a simple explanation for why transfer may fail.

Taken together, these metrics separate interpretive instability from predictive instability. That distinction is central to the paper, because a model may remain useful for prediction even when its coefficients are not stable enough for interpretation.

3 Simulation Study

This section will present the four main simulation scenarios, summarize the outputs, and describe the core figures and tables.

4 Real-Data Case Studies

We now report the first real-data case study using NHANES, with subgrouping by age decade. The analysis uses complete cases for $\{\text{AgeDecade}, \text{BMI}, \text{Gender}, \text{Poverty}, \text{DirectChol}, \text{Pulse}, \text{BPSysAve}\}$, leaving $n = 7116$ observations across $G = 8$ groups (0–9: 186, 10–19: 1080, 20–29: 1126, 30–39: 1138, 40–49: 1217, 50–59: 1116, 60–69: 782, 70+: 471).

The pooled model is

$$\text{BMI} \sim \text{Gender} + \text{Poverty} + \text{DirectChol} + \text{Pulse} + \text{BPSysAve},$$

and we fit the same formula separately within each age decade. We then compute sign stability, standardized drift, context-normalized drift, and predictor-wise objective components.

The aggregate stability summary indicates meaningful subgroup heterogeneity: mean sign stability is 0.814, mean standardized drift is 16.203, mean context-normalized drift is 11.339, and the objective score is 16.389.

Predictor-level metrics show that BPSysAve is the least stable term in this specification. Its sign stability rate is 0.571, standardized drift is 32.070, and context-normalized drift is 17.641. By contrast, Gendermale and DirectChol have perfect sign stability ($\text{SSR} = 1.0$), although DirectChol still shows nontrivial magnitude drift.

To visualize coefficient heterogeneity, we generated pooled and subgroup coefficient tables and fitted-curve plots, and added a dedicated BPSysAve coefficient plot with 95% confidence intervals by age decade. The BPSysAve plot shows a clear decline across age, with sign reversal in the oldest decades.

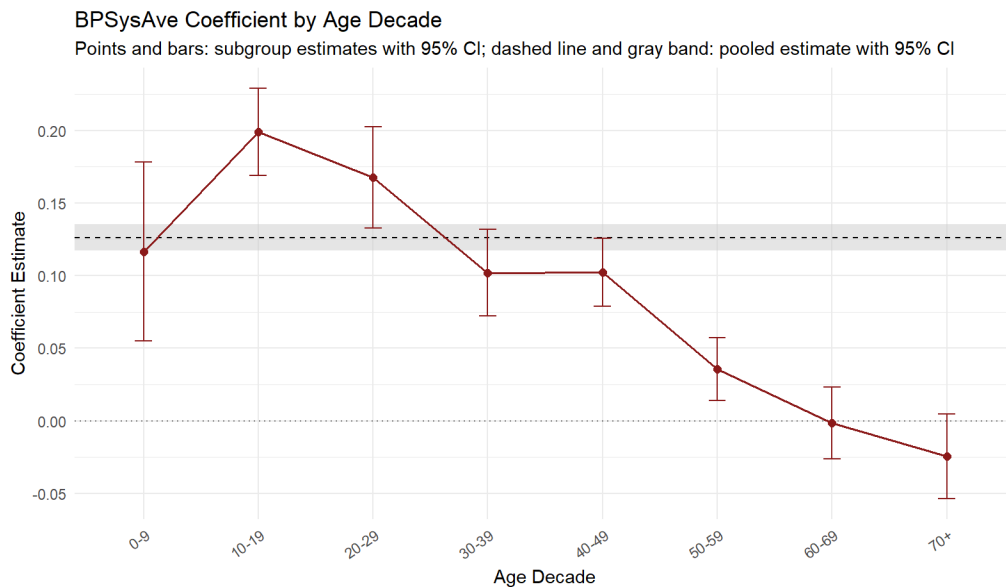


Figure 1: Estimated BPSysAve coefficient by age decade with 95% confidence intervals. The dashed horizontal line and shaded band show the pooled estimate and its 95% interval.

Because the youngest decade has a smaller sample ($n = 186$), we ran a size-sensitivity check. Excluding 0–9 reduces several drift values, but does not remove instability for BPSysAve (drift falls from 32.070 to 28.696, SSR changes from 0.571 to 0.524). Among similarly sized middle-age groups (10–19 through 50–59), BPSysAve sign agreement is high, while older decades still drive directional

mismatch. This pattern supports a regime-change interpretation rather than a pure small-sample artifact.

We also compared pooled versus age-specific predictive accuracy within each decade. Group-specific models improved RMSE and R^2 in every decade. The largest pooled failure occurred in 0–9 (pooled $R^2 = -3.485$ vs subgroup $R^2 = 0.241$), while notable gains remained in older groups (for 70+: pooled $R^2 = -0.086$ vs subgroup $R^2 = 0.130$). These results are consistent with coefficient instability and indicate that one pooled linear rule is not uniformly accurate across age strata.

5 Discussion

The NHANES age-decade case study reinforces the simulation message: pooled regression can mask meaningful subgroup heterogeneity even when average fit appears acceptable. In this application, instability is not only a magnitude effect but also a directional effect for selected predictors.

The strongest example is BPSysAve. Its coefficient drifts substantially across decades and crosses zero in older groups, while context-normalized drift remains large after adjusting for mean covariate separation. This indicates that the observed instability is not explained only by distributional shift in predictors.

A small-sample concern is reasonable for the 0–9 subgroup, and sensitivity checks confirm that this group contributes to instability for some terms. However, the main BPSysAve instability persists after excluding 0–9, and prediction gains from subgroup-specific models remain across all decades. The substantive conclusion is that age-stratified structure matters for both interpretation and prediction in this specification.

These findings should be interpreted as an empirical demonstration rather than a universal biomedical claim. The analysis uses one linear specification and in-sample subgroup comparisons. A natural next step is decade-wise cross-validation and bootstrap uncertainty quantification for drift metrics.

6 Conclusion

This section will provide a short closing summary of the paper’s main message.